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Article

Personalizing lung cancer screening recommendations for heterogeneous populations: a microsimulation study

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Abstract

Background: There is little guidance on how to personalize recommendations for lung cancer screening that accounts for the variation in expected net benefit from screening. We sought to explore the individual and population implications of identifying net benefit thresholds where lung cancer screening could be encouraged, discouraged, or offered as an option through neutral shared decision making due to screening being highly preference sensitive.

Methods: With a simulated US population of 40- to 80-year-old individuals who had ever smoked, we used microsimulation to estimate individualized quality-adjusted life-years saved with lung cancer screening. We then identified 2 net benefit thresholds for lung cancer screening that account for a range of patient preferences and scientific uncertainties and compared this approach with the current United States Preventive Services Task Force (USPSTF) guidelines.

Results: Our simulated population included 59 million people. In total, 15 million were USPSTF eligible; of those, 53% (8 million) maintained net benefit, even after accounting for unfavorable preferences about screening. Of the USPSTF population, 3% (450 000) are considered low net benefit (routinely discourage) and 47% (7 million) are in an intermediate gray area, where net benefit depends on patient preferences about lung cancer screening (offer-neutral shared decision making). Among adults who had ever smoked, 2.5 million are high net benefit but excluded by current USPSTF criteria; 20.5 million US adults who had ever smoked are intermediate net benefit but currently excluded by USPSTF criteria.

Conclusions: We estimate that half of the USPSTF lung cancer screening–eligible population is in a high-net-benefit group where lung cancer screening could be routinely encouraged. Current lung cancer screening eligibility criteria likely exclude many ever-smokers who are high net benefit and many more with intermediate net benefit.

Introduction

Based on 2 large, randomized trials,[1,2] the United States Preventive Services Task Force (USPSTF) concluded that annual lung cancer screening with low-dose computed tomography (CT) has, on average, moderate net benefit for persons aged 50-80 years with a 20 or more pack-year history and who still smoke or had quit within the past 15 years.[3] Such an eligibility threshold makes clinical implementation appear simple, but it ignores a key reality of clinical medicine: average benefit is a poor estimate for many patients.[4] In fact, all patients who smoke exist along a spectrum of net benefit from lung cancer screening that runs from low to high net benefit, leaving many in an intermediate gray area where the decision to undergo lung cancer screening is not clear-cut and a patient's informed preferences will determine whether screening is desirable or undesirable (ie, a "preference-sensitive" zone) (Figure 1).[5] For individuals faced with a preference-sensitive decision, experts advocate for a neutral approach to shared decision making.[6] Currently, the USPSTF universally recommends and Medicare requires shared decision making for all patients before lung cancer screening.[7]

A broad recommendation for a neutral approach to shared decision making is, however, most appropriate when the target population is homogeneous in its estimated net benefit. Spending precious clinical time on detailed shared decision making is of highest priority when the expected net benefit is preference sensitive (Supplementary Methods).[6,8] Not all interventions

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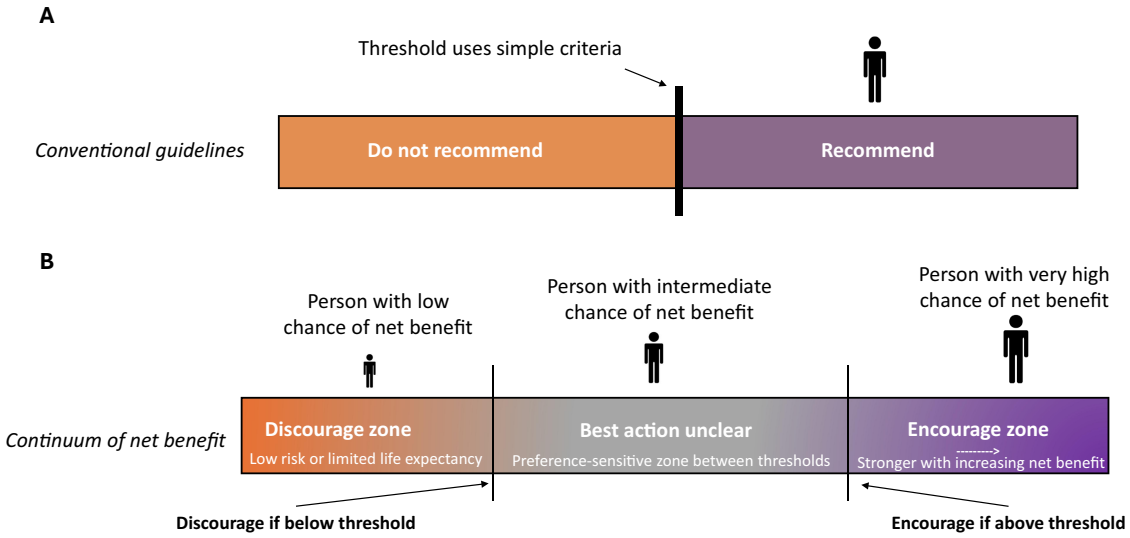


Figure 1. Conventional vs tailored guideline recommendations across the spectrum of individualized net benefit. (**A**) Conventional approaches to guideline recommendations typically use simple criteria, often based on the average net benefit of a given population and traditionally defined using the inclusion criteria of a randomized controlled trial. This approach produces a recommendation that appears dichotomous in nature. (**B**) Owing to the continuous nature of individualized risk, all patients exist on a spectrum of net benefit. Using a multivariable risk prediction tool that accounts for individualized lung cancer risk and life expectancy enables us to better understand the distribution of net benefit in a population and can enable the establishment of 2 thresholds, thus identifying a gray zone where net benefit is sensitive to a range of patient preferences and scientific uncertainties.

in medicine are highly preference sensitive, and even for a single intervention, the decision may be preference sensitive for some but not others.[9,10] It is well established that among patients eligible for lung cancer screening, estimated net benefit varies up to 10-fold due to substantial variation in lung cancer risk and life expectancy.[11] Thus, some eligible patients will have a high likelihood of net benefit, and a neutral discussion about lung cancer screening would be less appropriate for such high-net-benefit persons.

Clinicians and patients must ultimately decide whether to be screened. So, how might we help clinicians deal with the continuum of estimated lung cancer risk and life expectancy, to provide more individualized recommendations? For guidance to be useful, thresholds are needed, but a single threshold seems inadequate in a highly diverse population. In this article, we use an established microsimulation model to describe a conceptual framework and systematic approach to providing such guidance, identifying 2 thresholds to define 3 zones: an encourage (recommend) zone; a discourage (recommend against) zone; and an intermediate (preference-sensitive) gray zone, where the informed preferences of a patient are needed to know which course of action is best (Figure 1).

Methods

Study overview and model description

We used a microsimulation model for lung cancer screening to assess the distribution of individualized net benefit from 3 rounds of annual screening. Microsimulation is a type of decision analysis technique that models individual clinical pathways and can better capture the variation in patient characteristics and risk at baseline.[12] This particular microsimulation model has been used and described in detail elsewhere.[11,13] Briefly, it has a natural history component to simulate lung cancer outcomes in screening's absence and a screening component based on the National Lung Screening Trial[2] and a recent systematic review.[14] Updates to the model from prior publications and rationale for certain modelling decisions are in the Supplementary Methods.

Because we were interested in identifying an intermediate gray zone where differences in net benefit and net harm are sensitive to patient preferences, we conducted a sensitivity analysis as our primary analysis by varying assumptions for the amount of disutility (or degree of dislike) assigned to lung cancer screening and its downstream consequences. This process involved assessing outcomes for 2 different scenarios: (1) a favorable disutility scenario representing patients with a low degree of dislike for screening and its downsides and (2) an unfavorable disutility scenario representing patients with a high degree of dislike. Other parameters of the model were left at their base case assumption from prior studies.[11] The time horizon was lifetime, and the primary outcome was quality-adjusted life-years (QALYs). A discount rate of 3% was applied, and all analyses were done using R statistical software (R Foundation for Statistical Computing). This study was determined to be exempt by the VA Ann Arbor Healthcare System Institutional Review Board.

Study population and risk stratification

The study population was a nationally representative simulated cohort of adults who had ever smoked aged 40-80 years using data from the National Health Interview Survey. We chose to examine adults who had ever smoked in the United States to understand the variation in lung cancer risk and benefit from lung cancer screening in a broader but still at-risk population and compare this information with the USPSTF population. The National Health Interview Survey is an annual cross-sectional survey that collects self-reported data from noninstitutionalized adults. Sampling weights and covariance among variables were maintained when scaling to the US population.

All patients in the population were stratified by a well-validated, life-gain prediction model for lung cancer screening called Life-Years From Screening–CT (LYFS-CT).[15] We chose this prediction model for lung cancer screening in our analysis because it includes prediction of lung cancer death risk as well as

life expectancy for smoking adults, both of which are key determinants of potential life-year gains from lung cancer screening.[15] Other prediction models were also examined (Table S1).

Incorporating patient preference to define thresholds

Disutility estimates were assigned based on expert opinion of the modeling team informed by a systematic review of values and subsequent lung cancer screening models (Table S2).[16] The values in the review were estimated by various methods and included expert opinions of the authors of the studies included in the review. For some values, there were no corresponding values in the literature, and the expert opinion of the current modeling team was used to assign reasonable values based on available information. For example, no utility values reported for chest CT were found in the literature. Although disutility estimates derived from informed patients are ideal, available studies in lung cancer screening have not systematically captured all the disutility information for the model and do not routinely report the distribution of the estimates.

We defined the encourage threshold as occurring where, even despite highly unfavorable preferences toward screening, there is still net benefit or positive QALYs. This finding aligns with conceptual definitions of when a decision is no longer preference sensitive.[8] Next, we defined the discourage threshold as occurring where, even despite favorable preferences toward screening overall and only a modest disutility for the screening test itself, there is net harm or negative QALYs. The rationale for this approach is that if a patient cannot be expected to derive benefit from lung cancer screening after accounting for a modest disutility of the screening test itself, then those patients ought to be discouraged from screening.

Specifically, these thresholds were identified by locating where a locally estimated scatterplot smoothing curve fitted to the model outputs crossed zero on a graph plotting life-years gained against QALY gains for each disutility scenario described above. We then examined the data points for this intersection for consistency above or below zero to identify the encourage and discourage thresholds, respectively.

Target populations and outcomes

After encourage and discourage thresholds had been defined, we explored the implications of using these thresholds by identifying 5 key groups: (1) patients eligible by USPSTF criteria who were also high net benefit (above the encourage threshold), (2) patients who were high net benefit but missed by current USPSTF eligibility criteria, (3) patients who were low net benefit (below the discourage threshold) but included by USPSTF, (4) patients who were intermediate net benefit (gray area between the discourage and encourage thresholds) but missed by current USPSTF criteria, and (5) patients who were USPSTF eligible and intermediate net benefit. We chose to highlight race and ethnicity in our results to examine the proportion of these groups included and excluded by the current status quo. Inclusion of race and ethnicity has been shown to be important for lung cancer screening considerations.[17]

To show the consequences of screening with these decision thresholds, we examined key outcomes and compared them with screening decisions using the 2021 USPSTF lung cancer screening criteria. Key outcomes included lung cancer deaths prevented per 1000 persons, life-years gained per 1000 persons, days of life gained per person screen, and QALYs gained per 1000 persons. Model parameters were set to the base case for these estimates.

Results

Characteristics of the simulated population are reported in Table S3. Figure 2 shows the distribution of predicted life-days gained within the USPSTF-eligible population and the entire study population of adults who had ever smoked aged 40-80 years.

Under current USPSTF guidelines, 15 165 433 persons in the study population of adults who had ever smoked and were 40 to 80 years of age would be eligible for lung cancer screening (25.7% of the study population). If everyone in this group were to undergo lung cancer screening, we estimate 5.8 lung cancer deaths per 1000 screened persons would be prevented, 93.3 life-years per 1000 persons would be gained, 21.6 days of life would be gained per person screened, and 27.0 QALYs would be gained per 1000 persons screened.

Encourage threshold

We identified the encourage threshold at 16.28 life-days gained, which represents the 85th percentile of risk in the study population (Figure 3). Overall, 10 383 194 persons in the study population (17.6%) were in the encourage zone. In Tables S4 and S5, we present results using an alternative, lower disutility for the screening test (–2 days for every year screened) to define the encourage threshold.

Included by USPSTF and above the encourage threshold

Among individuals eligible for lung cancer screening by USPSTF guidelines, 52% (about 8 million) were identified above the encourage threshold (Table 1). Compared with the population included by USPSTF but below this threshold, 75% of this high-net-benefit population is aged 60 years or older (vs 39%) and 77% have smoked for more than 40 years (vs 33%) (Table 1). If everyone in this group were to undergo lung cancer screening, we estimate 9 lung cancer deaths per 1000 persons screened would be prevented, 132 life-years per 1000 persons would be gained, 33 days of life would be gained per person screened, and 38 QALYs would be gained per 1000 persons screened (Table 2).

Above encourage threshold but excluded by USPSTF

We estimate 2.5 million persons (4.3% of the study population) in the United States are above the encourage threshold but excluded by current USPSTF criteria. When we assessed the characteristics of this population, we found that 25% were non-Hispanic Black compared with 7.7% of the current USPSTF-eligible population under the encourage threshold (Table 1). Adding this population to the current non-Hispanic Black population eligible by USPSTF criteria would result in a 54% increase in the size of the lung cancer screening–eligible Black population. Among the high-net-benefit population currently excluded by USPSTF criteria, about 50 000 (1.9%) are aged 40-49 years, and 32.8% quit smoking more than 15 years ago; 85% were 60 years of age or older compared with 38.6% for persons USPTSF eligible but below the encourage threshold (Table 1).

If everyone in this group were to undergo lung cancer screening, about 6 lung cancer deaths per 1000 persons screened would be prevented, 84 life-years per 1000 persons would be gained, 23 days of life would be gained per person screened, and 22 QALYs would be gained per 1000 persons screened (Table 2).

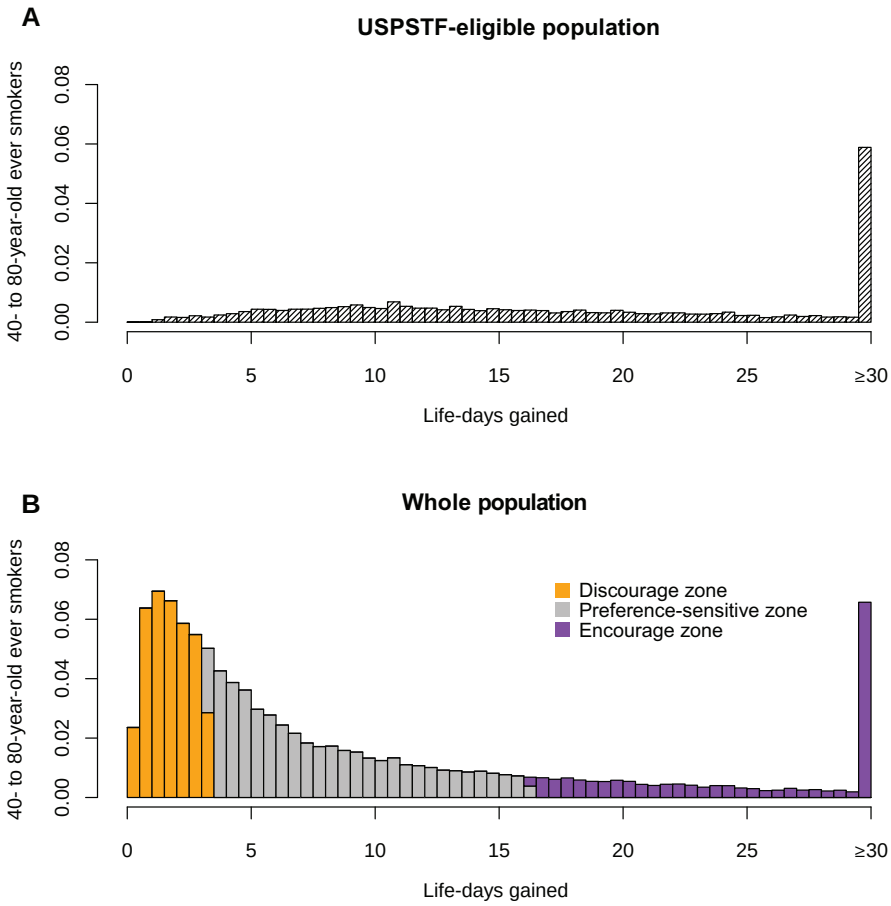


Figure 2. Distribution of life-days gained within the USPSTF-eligible population and the 40- to 80 year-old ever-smoking population. (**A**) Distribution of risk profiles across the entire USPSTF-eligible population. The "≥30" bar on the far right of the x-axis represents all persons at or above 30 life-days gained (93th percentile, about 6% of the USPSTF-eligible population). (**B**) Distribution of risk profiles across the entire population of 40- to 80-year-old adults who had ever smoked. Colored bars represent patients within the discourage zone, preference-sensitive zone, and encourage zone. The "≥30" bar on the far right of the x-axis represents all persons at or above 30 life-days gained (93th percentile, about 7% of the entire study population). USPSTF = US Preventive Services Task Force.

Discourage threshold

We identified the discourage threshold at 3.28 life-days gained, representing the 33rd percentile of risk in the study population (Figure 3). More than 20 million persons in the study population (36.5%) are in the discourage zone. Table 2 shows the population outcomes of screening in this group.

Below discourage threshold, included by USPSTF

Of the individuals in the discourage zone, 2.8% (430 035 people) are USPSTF eligible; 98% of the persons in this group were aged 50-59 years, and almost one-quarter were in the first and second deciles of risk (Table 1). If everyone in this group were to undergo lung cancer screening, fewer than 1 lung cancer death per 1000 persons screened would be prevented, about 18 life-years per 1000 persons would be gained, 2 days of life would be gained per person screened, and 3 QALYs would be gained per 1000 persons screened (Table 2).

Intermediate, preference-sensitive zone

In all, 27 318 411 people in the study population were in the intermediate, preference-sensitive gray zone. Almost 7 million of those persons in the preference-sensitive zone were USPSTF eligible, which is 44.8% of the current USPSTF-eligible population.

Intermediate, preference-sensitive zone but excluded by USPSTF

About 20.5 million adults who had ever smoked in the preference-sensitive zone are currently excluded by USPSTF guidelines, which comprises 35% of the entire study population (Table 2). About 13% of that population is non-Hispanic Black compared with the 7.7% in the current USPSTF-eligible population, which, if added to the current USPSTF-eligible population, would more than double the non-Hispanic Black population (225% increase). Sixteen percent of this population is aged 40-49 years, 56% are 60 years of age or older, and approximately 50% had quit smoking more than 15 years previously. Outcomes for this group are presented in Table 2.

In Tables S6 and S7, we also present the population characteristics and outcomes for the entire population above the lower discourage threshold, which includes both intermediate (preference-sensitive) gray zone and encourage zone individuals.

Discussion

We present a conceptual framework using microsimulation to identify an intermediate gray zone for lung cancer screening by setting 2 thresholds: 1 below which screening is discouraged and another above which screening is encouraged. We estimated that 52% of the current USPSTF-eligible population can expect net

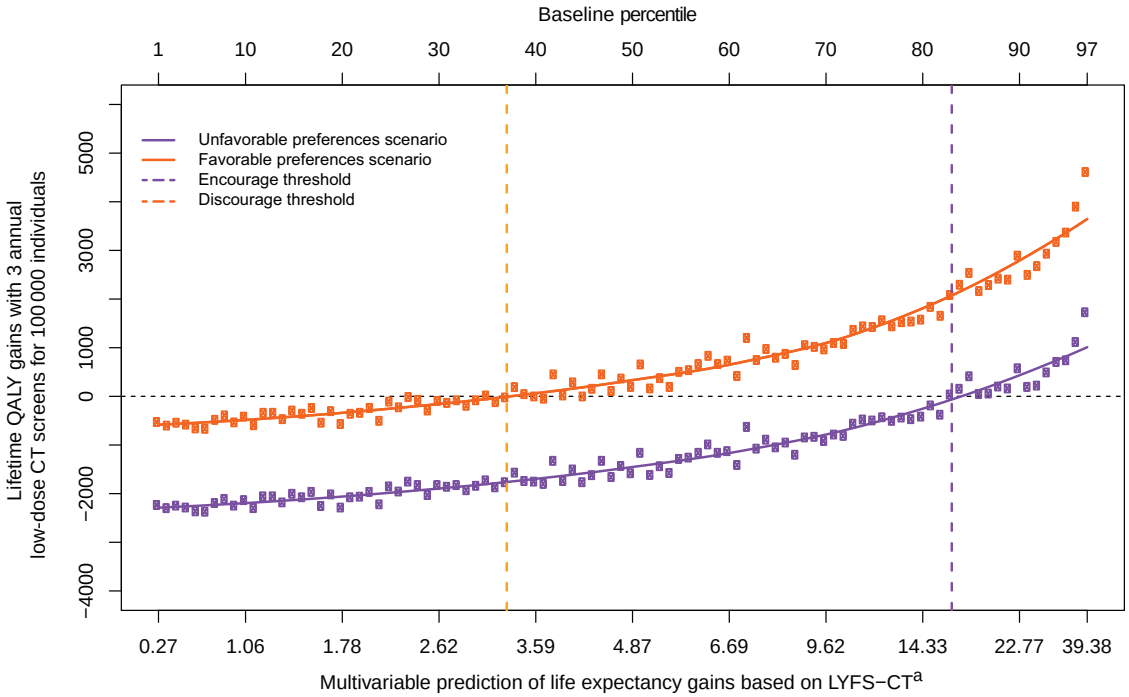


Figure 3. Establishing encourage and discourage thresholds among 40- to 80-year-old US adults who have ever smoked. Sussman plot demonstrates locally estimated scatterplot smoothing curves and data points of the favorable preferences (low disutility) scenario and unfavorable preferences (high disutility) scenario. Vertical dashed lines represent discourage (left) and encourage (right) thresholds where the associated curves cross zero. Between vertical dashed lines lies the intermediate, preference-sensitive zone; left of the discourage threshold is the discourage zone; and to the right of the encourage threshold is the encourage zone. CT = computed tomography; LYFS-CT = Life-Years From Screening–Computed Tomography; QALY = quality-adjusted life-year.

*The LYFS-CT calculates life-years gained from lung cancer screening using the difference between estimated life expectancy without screening and estimated life expectancy with 3 annual CT screens and 5 years of follow-up. This model incorporates a validated, individualized, all-cause mortality model, the Lung Cancer Death Risk Assessment Tool, and relative risk reduction estimates from the National Lung Screening Trial.[15]

benefit from lung cancer screening, even after accounting for unfavorable preferences about lung cancer screening. Despite present recommendations to conduct shared decision making for all individuals eligible for lung cancer screening, such screening is *less* preference sensitive for this high-net-benefit group; thus, neutral shared decision making may be less appropriate. Based on our analysis, current guidelines do well in excluding many low-risk, low-net-benefit patients, but they also exclude many persons with appreciable lung cancer risk and reasonable life expectancy who have at least moderate net benefit: more than 2 million adults who are high net benefit and more than 20 million adults at intermediate net benefit for whom lung cancer screening may be reasonable if it aligns with their preferences. Disproportionately, individuals in these intermediate- or high-net-benefit groups who are excluded by current criteria are Black, are older (≥ 60 years), have smoked for a longer duration but with less intensity (<20 pack-years), and quit more than 15 years previously.

Traditional approaches to producing guideline recommendations[18,19] typically focus on setting a single threshold for a specific population that creates 2 seemingly distinct groups: individuals for whom the intervention should be recommended (weakly or strongly) and individuals for whom it should not. Such simple criteria can quickly communicate to busy clinicians which broad populations are likely to benefit, on average. We found, however, that simple criteria may have considerable limitations for guiding lung cancer screening decisions for individual patients with variable lung cancer risk and life expectancy and that simple criteria likely exclude many individuals who have a substantial chance of benefitting from lung cancer screening and many more for whom lung cancer screening could be reasonable. Setting 2 risk thresholds as opposed to 1 enables us to better address the continuous nature of net benefit from lung cancer screening, account for a range of patient preferences, and identify persons for whom the decision to undergo lung cancer screening is most preference sensitive.

Although individualized risk prediction has the potential to introduce greater complexity into lung cancer screening recommendations and decisions, the 2-threshold approach can still clearly communicate a recommended course of action to clinicians: Patients in the encourage zone could be routinely encouraged to undergo lung cancer screening, and persons in the preference-sensitive zone could undergo a neutral shared decision making process to further elucidate patient preferences and determine the best course of action, with persons in the discourage zone being ineligible.

This approach also provides a path for improving the delivery of interventions to promote lung cancer screening uptake.[20] Lung cancer screening rates are lower in the United States than are other US cancer screening programs,[21] and persons in the encourage zone could be prioritized for effective but limited interventions known to increase screening, such as screening coordinators and ride programs.[22] Related research has shown that an individualized, net benefit–based approach to lung cancer screening can increase uptake and close important care gaps.[23-25]

We identified large populations of adults who had ever smoked and may benefit from lung cancer screening but are currently excluded by USPSTF criteria. We estimated that persons excluded but above our encourage threshold would have more

Table 1. Characteristics of overlapping and nonoverlapping populations using simple (USPSTF) vs multivariable criteria.

	Above encourage, not USPSTF	Below encourage, meet USPSTF	Above encourage, meet USPSTF	Below discourage, meet USPSTF	Below discourage, not USPSTF	Intermediate, meet USPSTF	Intermediate, not USPSTF
Total, No.	2 528 786	7 311 026	7 854 408	430 035	21 673 615	6 880 990	20 437 420
Race and ethnicity, %							
American Indian or Alaska Native	0.9	1.3	0.6	3.4	1.3	1.2	0.8
Asian or Pacific Islander	1.9	3.2	1.2	1.2	4.5	3.3	2.5
Hispanic	3.5	9.0	2.5	17.1	14.6	8.5	7.8
Non-Hispanic Black	24.9	5.6	9.7	4.4	6.2	5.7	12.9
Non-Hispanic White	68.8	81.0	86.0	73.8	73.3	81.4	76.0
Sex, %							
Male	53.2	57.1	57.5	60.1	52.3	56.9	53.7
Female	46.8	42.9	42.5	39.9	47.7	43.1	46.3
Age group, %							
40-44 y	0	0	0	0	29.6	0	3.2
45-49 y	1.9	0	0	0	20.5	0	13.0
50-54 y	1.2	32.7	6.3	64.5	15.0	30.8	12.5
55-59 y	7.8	28.5	16.9	33.4	13.7	28.2	12.9
60-64 y	14.6	20.8	24.6	1.1	10.2	22.0	14.1
65-69 y	20.0	11.1	22.6	0.1	6.5	11.8	15.8
70-74 y	26.5	5.1	17.9	0.8	3.1	5.4	15.2
75-79 y	24.3	1.6	10.3	0.2	1.2	1.7	11.1
Comorbidities, %							
0	51.0	52.7	49.4	55.0	73.5	52.5	58.5
1	20.5	17.4	19.5	17.4	13.2	17.4	18.5
2	12.5	11.2	13.2	7.7	6.6	11.5	10.5
3	7.7	7.6	8.2	8.2	3.2	7.6	6.3
≥4	8.3	11.1	9.6	11.6	3.6	11.1	6.3
Lung Cancer Death Risk Assessment Tool decile, %							
1	0	0.1	0	1.1	27.4	0	0
2	0	1.3	0	21.7	27.4	0	0
3	0	2.9	0	49.1	26.7	0	0.8
4	0	4.7	0	22.4	13.6	3.6	13.4
5	0	8.8	0	4.5	3.7	9.1	22.0
6	0	15.5	0	0.3	1.0	16.5	22.5
7	0.2	26.6	0	0.9	0.2	28.2	19.3
8	8.9	28.3	8.4	0	0.04	30.1	14.5
9	43.4	10.0	34.9	0	0	10.6	6.7
10	47.5	1.8	56.7	0	0	1.9	0.7
Smoking duration, %							
0-20 y	1.9	1.5	0.07	9.6	67.7	1.0	26.4
21-30 y	11.8	17.4	1.4	77.0	28.4	13.7	31.3
31-40 y	29.2	48.3	17.8	10.9	3.8	50.6	33.8
41-50 y	32.6	29.6	51.6	1.9	0.06	31.4	7.5
51-60 y	20.9	3.1	25.0	0.7	0.01	3.3	0.9
Pack-year history, %							
0-9	29.0	0	0	0	62.1	0	36.0
10-19	35.7	0	0	0	23.7	0	34.2
20-29	2.6	43.0	12.9	90.0	12.1	40.1	13.0
30-39	7.1	33.8	17.1	6.0	1.3	35.1	9.8
40-49	6.5	11.5	27.5	1.7	0.5	12.1	3.1
50-59	5.5	3.6	16.2	1.9	0.2	3.8	1.5
≥60	13.5	8.6	26.2	0.4	0.2	9.1	2.3
Years since quitting, %							
0-5	65.7	64.0	86.5	18.6	28.8	66.8	43.3
6-10	1.3	20.1	8.3	33.0	10.3	19.3	4.5
11-15	0.3	16.0	5.2	48.4	9.4	13.9	3.6
16-20	15.0	0	0	0	12.0	0	10.3
21-25	6.8	0	0	0	8.0	0	7.2
26-30	7.3	0	0	0	10.6	0	10.2
>30	3.7	0	0	0	20.8	0	21.0

Abbreviation: USPSTF = US Preventive Services Task Force.

lung cancer deaths prevented and QALY gains per 1000 persons screened than many who are currently eligible by USPSTF criteria. Further, many more who are excluded by USPSTF criteria but in our intermediate zone had more estimated lung cancer deaths prevented and QALY gains per 1000 persons screened than some persons currently meeting USPSTF eligibility. These striking results suggest that current criteria result in substantial inequities of care and contradict the equal management of equal risk principle.[26]

These results are similar to those from a recent study that also identified a high-net-benefit group for lung cancer screening using fast-and-frugal trees.[27] As described in that article and

Table 2. Population outcomes of overlapping and nonoverlapping groups.

	Above encourage, not USPSTF	Below encourage, meet USPSTF	Above encourage, meet USPSTF	Below discourage, meet USPSTF	Below discourage, not USPSTF	Intermediate, meet USPSTF	Intermediate, not USPSTF
Total, No.	2 528 786	7 311 026	7 854 408	430 035	21 673 615	6 880 990	20 437 420
Lung cancer deaths prevented per 1000 persons screened	5.9	2.2	9.1	0.6	0.3	2.3	1.4
Life-years gained per 1000 persons screened	83.9	51.9	131.8	18.2	12.9	54.0	37.1
Days of life gained per person screened[a]	23.2	9.6	32.8	2.2	1.7	10.1	6.9
QALYs gained per 1000 persons screened	21.9	15.0	38.2	2.8	0.7	15.7	9.6

Abbreviations: QALY = quality-adjusted life year; USPSTF = US Preventive Services Task Force.
[a] Estimate obtained from the R *lcrisk* package, not the result of the microsimulation model; thus, the relationship between this estimate and the rest of the rows in the table is not scalar.

confirmed here, the differences in populations captured by USPSTF criteria and the prediction model likely have to do with the USPSTF's use of pack-years as a surrogate for cumulative smoking exposure vs the LYFS-CT model, which weighs duration of tobacco use separately from smoking intensity. Smoking duration seems to be a more considerable risk factor for lung cancer that should be considered separately from smoking intensity.[28,29]

We estimate that more than 20 million adults who had ever smoked and are between 40 and 80 years of age fall within the intermediate gray zone but are excluded by current criteria. Adding this population to the current USPSTF-eligible population would more than double the current population eligible for lung cancer screening, meaning that there are many more patients than currently eligible who would potentially benefit from being offered lung cancer screening through a neutral shared decision making discussion. Opening eligibility to a broader population of former smokers is consistent with new American Cancer Society guidelines that removed the years since quitting criterion.[30]

In practice, identifying more people for whom lung cancer screening may be reasonable could present substantial time allocation problems to already busy clinicians. Proposals for more efficient ways to provide shared decision making are available,[31] and the delivery of patient decision aids before an appointment would be essential to effective shared decision making. Nonetheless, adding this group would substantially and disproportionately increase the number of non-Hispanic, Black patients eligible for lung cancer screening, which may provide a mechanism for decreasing important lung cancer disparities known to exist among this population.[32] Prior studies have shown that the 2021 USPSTF lung cancer screening criteria could potentially reduce racial and ethnic disparities in lung cancer outcomes,[33,34] but our analyses align with more recent studies that suggest an even greater potential to reduce these disparities through a prediction-based approach.[17,27,35] Further work to improve lung cancer screening uptake and adherence among these high-risk groups is needed.

Our study has several limitations. Setting thresholds is ultimately a value judgment. Simulation and prediction can provide useful information to decision makers, but models have their own uncertainties and imprecision regarding input parameters and risk model calibration error.[11] Setting the range of patient disutility estimates for the burden and complications for any type of screening is difficult, but further research to better estimate these disutilities is needed.[36-39] We tried to err on the side of a broad range of individual patient disutilities; thus, we may have overestimated the width of the preference-sensitive zone. The thresholds we present should be viewed as rough guideposts.

In conclusion, simple eligibility criteria for lung cancer screening exclude many patients with higher-than-average net benefit from lung cancer screening and exclude even more for whom the decision to undergo lung cancer screening is likely preference sensitive. More than half of currently eligible patients could receive stronger encouragement to undergo lung cancer screening rather than neutral shared decision making. To realize these potential benefits of lung cancer screening, greater resources and better strategies and interventions are needed to improve the uptake of screening among persons at risk.

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Author contributions

Joshua B. Rager (Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing—original draft), Pianpian Cao (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Writing—review & editing), Rodney A. Hayward (Conceptualization, Investigation, Methodology, Supervision, Writing—review & editing), Rafael Meza (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Writing—review & editing), Hormuzd A. Katki (Methodology, Resources, Software, Supervision, Writing—review & editing), Jeremy B. Sussman (Conceptualization, Methodology, Supervision, Writing—review & editing), and Tanner J. Caverly (Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Visualization, Writing—original draft, Writing—review & editing)

Supplementary material

Supplementary material is available at *JNCI: Journal of the National Cancer Institute* online.

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Conflicts of interest

T.J.C. reports an open source Apache 2.0 license for a freely available decision support tool (screenlc.com) for which he does not receive compensation. H.A.K., who is a *JNCI* editorial board member and co-author of this article, was not involved in the editorial review or decision to publish the manuscript.

Data availability

The patient and population data used for this study are available publicly through the National Health Interview Survey at cdc.gov. The microsimulation model is not publicly available; for those interested in this model, please visit cisnet.cancer.gov.

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